

From Code to Careers: Linking Open-Source Software Signals to AI Skill Demand in the Labour Market

Dafydd James Thomas

Department of Computer Science
Aberystwyth University
Wales

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Abstract

The abstract stands alone as a very short version of the dissertation.

The abstract should state the scope and principal objectives of the project, describe the methods, summarize the results and state the principal conclusions.
(Max. 500 words.)

Declaration of originality

I confirm that:

- This submission is my own work, except where clearly indicated.
- I understand that there are severe penalties for Unacceptable Academic Practice, which can lead to loss of marks or even the withholding of a degree.
- I have read the regulations on Unacceptable Academic Practice from the University's Academic Quality and Records Office (AQRO) and the relevant sections of the current Student Handbook of the Department of Computer Science.
- In submitting this work, I understand and agree to abide by the University's regulations governing these issues.

Name: Dafydd James Thomas

Date: 05/09/2025

Consent to share this work

- By including my name below, I hereby agree to this thesis being made available to other students and academic staff of the Department of Computer Science, Aberystwyth University.

Name: Dafydd James Thomas

Date: 05/09/2025

Acknowledgment

To whoever has the patience to read this :-)

This section is customary, but not obligatory. It makes a brief statement of thanks to those who have helped.

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Chapter 1

Introduction

In recent times, we are seeing that skills and job requirements are evolving rapidly. As the culmination of various effects, including the development of new technologies and evolving consumer preferences, the labour force must adapt in order to satisfy the new requirements. Given the historic pace of labour demand shifts, traditional methods of forecasting these changes are not well-equipped to handle this blistering, and accelerating, rate of change.

By these so-called "Traditional" methods, I mean that which accompany government or firm-level analysis of the skills market. These depend upon data which are limited in the following ways:

1. They use structured data
2. They collect the data periodically (e.g. Annually / Quarterly)

An important feature of these methods is their approach to modelling the dynamics influencing labour demand. Of key importance is the impact of innovation, and how new technology is estimated to affect skill requirements.

This paper aims to investigate whether Open Source Software (OSS) innovation Signals, derived from unstructured text data, can meaningfully contribute to the forecast of AI-related skill requirements present in job postings.

To break this down, there are a few key ideas being explored:

1. Can open source software data represent information about Innovation in the AI sector?
2. Can AI-related skill requirements be modelled the textual data within job postings?
3. Can we integrate the two data channels to produce a quantified measure of their relationships?

Put simply, if we can measure how similar an OSS repository (innovation-source) is to job postings, in terms of the skill-profile and concepts involved, then we can represent how this innovation-source has diffused over time (i.e. by the date of job posting). By relating this to the temporal development of the innovation-source, as represented by some other measure of the OSS repository’s history, we can the development-diffusion timeline of that particular innovation.

If true, this insight has tangible benefits for the practice forecasting skill demand in the AI sector, within which open-source software is a prolific source of innovation. Moreover, job postings & OSS data are high-frequency and accessible, meaning that more granular and fine-tuned predictions can be produced in effective real-time.

1.1 Thesis Argument

As a student of philosophy, I find that a strong start to any paper is to frame the content and argument clearly.

The function of this is to provide a clear view of what is either assumed to be true, what is *proven* to be true (as follows from the relevant literature), or what I expect to *demonstrate*. Thereafter, the conclusion necessarily follows, and it can be seen that the function of this paper is to prove the hypotheses as so labelled.

Argument 1.1: Thesis Argument

Assumptions

- A1** Open-source activity reflects experimentation.
- A2** Repo stars are, in large numbers, informative.
- A3** Only English language sources of text are considered.
- A4** The location of a job posting has a negligible effect.

Hypotheses

- H1** GitHub activity precedes job market demand.
- H2** AI-related skill mentions are represented semantically in similar ways between Job Postings and GitHub repository text, enabling cross-domain semantic comparison.

Theoretical Premises

- P1** There exist validated NLP methods for extracting technical skills from job postings with high precision and recall [1].
- P2** Contextual embeddings (e.g., Sentence-BERT) map job postings and repository descriptions into a shared feature space where cosine similarity reflects semantic similarity, as validated on semantic textual similarity benchmarks **reimers-gurevych-2019-sentence**.
- P3** Technical skills and repository descriptions are expressed in relatively standardized terminology, enabling reliable extraction and semantic comparison [2].
- P4** Stars indicate repository visibility and perceived popularity, while forks indicate community engagement and experimentation with the repository [3].
- P5** Sustained repository activity (commits, issues, pull requests) signals ongoing experimentation and adoption of new technologies **borges-2016-popularity**.
- P6** Early engagement with repositories may reflect emerging technology trends that are later reflected in labor market demand.
- P7** Job postings represent explicit employer demand for skills and thus serve as a proxy for labor market demand trends.
- P8** Temporal trends in job postings correlate with broader industry hiring trends **radlinski-2011-learning**, [1].
- P9** Skill mentions in job postings are described with relatively consistent terminology, enabling robust extraction using NLP techniques [2].
- P10** Cosine similarity between embeddings in the same feature space provides a quantitative measure of semantic similarity suitable for ranking items **mikolov-2013-distributed**, **reimers-gurevych-2019-sentence**.
- P11** Cosine similarity preserves approximate local neighborhoods: if two items are both highly similar to a shared reference embedding, they are likely to be semantically related to each other **ethayarajh-2019-contextual**.
- P12** Cosine similarity induces a **p**artial ordering of job postings with respect to each repository: if $\cos(J_A, R) > \cos(J_B, R)$, then J_A is considered more semantically aligned with R than J_B , enabling

1.2 Aims & Objectives

1.2.1 Primary Aim

To develop and evaluate an AI-related skill demand forecasting methodology, integrating innovation signals (from Open-Source Software in particular) and using job ads as a proxy for skill demand.

1.2.2 Objectives

1. **Data Collection:** Collect & collate data for both channels in:
 - (a) Open-source Software (GitHub): *textual content & historic measure of development*
 - (b) Job Postings (e.g. LinkedIn, Indeed) - *content & date-posted*
2. **Representation:** Develop a high-quality representation method for integrating textual data from both channels, involving:
 - (a) **Build a Baseline:** A weak, keyword-based mechanism.
 - (b) **Advanced Representation:** An intelligent and robust method for representing text as embeddings.
 - (c) **Experimentation:** Fine tune and alter the components of embedding model, evaluating against the baseline.
3. **Time series:** Investigate Relationship of derived time series, looking at Granger causality values and Pearson correlations.
4. **Discuss:** Critically evaluate the derived results, With reflections on the benefits and limitations of the approach.

1.3 Roadmap

In the following chapter, I discuss related pieces of work and how their findings informed the paradigm for this project’s implementation. Given the domain intersectional nature of this paper, works that are ”related” include those on the topics of economic theory, natural language processing and AI itself.

Following this, the implementation of this project is detailed. Firstly, this concerns how data was acquired and preprocessed. Thereafter, the primary component of the pipeline is discussed - namely, the representation pipeline. Within this, I demonstrate both how the baseline model and the representation model are built,

discussing the involved components and how experimentation can be conducted to find optimal constructions. The subsequent section demonstrates the downstream task of these representation results - in the context of time series analysis, the findings of our representation pipeline are presented alongside the data of GitHub Stars. Having demonstrated the function of the entire pipeline, this next section therefore details the results and rationale of several sets of experiments.

What thereafter follows is a critical evaluation of the process, discussing the project's limitations and its successes, as well as the opportunities for future work. Finally, We discuss in the conclusion the outcomes and results of this paper, and their implications.

Chapter 2

Literature review

2.1 Introduction

Before embarking on implementing the project, extensive research was conducted to understand the various elements at play. The following depicts a list of questions that were devised in the research stage, and are answered thereafter. It is with this information that we can build a picture of what is supported (and what is overlooked) by existing research. This serves as a foundation for my work, giving precedent to the study of the thesis at hand.

As previously mentioned, the thesis argument in the Introduction Chapter grounds its premises on empirical grounds, according to the research thusly discussed. I shall make reference as appropriate to the premises in question.

1. AI Revolution & Labour shifts

- (a) What's happening with AI?
- (b) How does this compare to previous technological shifts?
- (c) How does this affect the workforce?

2. Labour Demand Analysis

- (a) How is labour demand forecasted?
- (b) Does AI pose a problem to these methods?
- (c) Are Job Postings useful here?

3. Innovation & Open-Source Software

- (a) What is Open Source software?
- (b) Is OSS a valuable innovation signal? Why?

- (c) How can we measure the adoption rate of OSS?

4. Methodological Techniques

- (a) How can we effectively represent semantic content quantitatively?
- (b) What are the considerations for comparing semantic representations of different domains?
- (c) To what extent can we infer relationships between two time-series?

2.2 Labour Demand Forecasting

2.2.1 Conceptual Overview

2.2.2 Empirical Findings and Applications

2.3 Innovation and Labour Demand Shifts

2.3.1 Theoretical Background

2.3.2 Empirical Evidence

2.4 Open Source Software (OSS) as a Signal of Innovation

2.4.1 OSS and Software Innovation

Open source software (OSS) is increasingly a rich ecosystem for The development of new software technologies. Often it is regarded as a hub for experimentation and innovation, and is a unique depiction of innovation outside of the incentive structure observed by private economic agents.

The authors of [4] investigate the question “Do weak ties facilitate creativity in software in similar ways?”, and find that a developer/contributor’s innovativeness is strongly correlated with the breadth and diversity of projects they engage with. This is highly suggestive of the power that the OSS ecosystem has to induce a positive externality-like effect in innovation circles.

[5] “identify four main patterns of popularity growth, which are derived after clustering the time series representing the number of stars of 2, 279 popular GitHub repositories”. Growth patterns identified are Slow, Moderate, Fast, and Viral.

[6] expand thereafter by:

- expand their analysis to 5000 repositories
- survey 791 developers to identify the motivation / reasons behind starring
- investigate of factors influencing star-growth (e.g. marketing)
- Develop a Machine Learning method to identify specific factors that predict a GH repository’s popularity growth.

[7] - “dynamics of innovation in OSS ecosystems are quite similar to the dynamics of innovation in patenting and scientific research”.

[8] - A report on the costs and necessary upkeep of the ecosystem of OSS, and how it should be treated as a public good.

2.4.2 GitHub Metrics

2.5 Job Postings as a Proxy for Labour Demand

2.5.1 Conceptual Basis

2.5.2 Empirical Support

2.6 NLP for Skill Extraction in Job Postings

2.6.1 Skill and Competence Taxonomies

To represent the “skill profile” of a Job Posting (JP) or a GitHub Repository (GHR), we need a **taxonomy** to define the concepts of a “skill”. Official classification bodies such as ESCO and O*NET exist for this broad purpose. They define hierarchical classifications to depict skills requirements for occupations.

The term *skill* itself is, however, only narrowly defined by the *European Skills, Competences, Qualifications and Occupations* (ESCO) classification [9] as “the ability to apply knowledge and use know-how to complete tasks and solve problems.” ESCO further distinguishes *knowledge* as “the outcome of the assimilation of information through learning,” encompassing facts, principles, theories, and practices related to a field of work or study, and defines *competences* as “the proven ability to use knowledge, skills, and personal, social, and/or methodological abilities in work or study situations and in professional and personal development.” Thus, competences capture the broader capacity to apply both knowledge and skills within specific contexts.

These classifications are problematic for the purposes of this project, because in practice, ESCO does not make a clear distinction between skills and competences

within its “skills pillar”. ESCO also defines many items that are ability-wise relevant to AI (e.g. specific programming languages) as knowledge rather than skills. Additionally, ESCO classifications are manually defined, and may therefore be missing the nuance of recent & emergent taxonomies in the domain of AI. The strict and somewhat outdated ESCO classifications are therefore problematic for framing what the “skills” in this domain are.

Therefore, for the remainder of this paper, I shall stipulate the following:

Stipulation

The term “skill” is used broadly to encompass all three ESCO categories insofar as they relate specifically to AI technology.

This serves to simplify the discussion of how GitHub and Job Postings relate within the domain of AI technology, and should therefore not be confused with how ESCO defines a skill.

2.6.2 Background NLP

The simplest Natural Language Processing (NLP) technique is keyword matching: querying a body of text to determine the presence of a defined keyword. Though straightforward, This method is too narrow to capture the complexity of *natural* language, which encompasses alternative terms of phrase, acronyms and so on. To be useful, this method would involve curating a large dictionary of terminology, which is laborious.

An improvement would be to use fuzzy matching techniques, which allows for matching on typos and some small patterns alternative phrasing. However, this approach remains limited in its ability to capture *meaning*. It is not enough to simply define a match on a word-for-word basis, because a crucial element of what makes natural language meaningful is the context. For example the word “Python” alone means something very different when seen in the context of “Python, cobra and anaconda” as compared to “Python, SQL and Java”.

Building on some fundamental breakthroughs in the field of NLP, the seminal paper [10] depict a *Transformer architecture* that represents natural language with a contextual appreciation for each word (or token) in a sentence. With this, we can “embed” a body of text high dimensional feature space that meaningfully captures the text’s contextual and semantic properties. This technique therefore improves computational efficiency and reduces the manual oversight needed to process and depict what a piece of natural language is alluding to.

2.6.3 NLP for Skill-extraction

In [2], the authors develop a data driven methodology to extract and represent a job-skill taxonomy. Emphasizing that they do so independently of official boards for taxonomy definition (e.g. O*NET and ESCO), they Leverage NLP and network analysis methods for an algorithmic methodology to construct skill taxonomies, which remain adaptable and avoid expert elicitation. The significance of this is that it reinforces the feasibility of using NLP techniques, and specifically sentence embeddings, to extract skills from job postings.

Named Entity Recognition (NER) is a method of Natural Language Processing which identifies and extracts known entities (such as companies, people, or locations) from a body of text. The authors of [1] develop a specialized Named Entity Recognition (NER) Model that identifies and extracts the skills from within the body of text for a job posting.

2.7 Sentence Embeddings and Cross-Domain Semantic Comparison

2.7.1 Overview of Sentence Embeddings

As previously mentioned, the development in the field of Natural Language Processing was the Transformer architecture, which enriches the embeddings of tokens in a sentence with contextual meaning. **Somehow** sentence transformers distill a sequence of enriched token embeddings into a single embedding. **HOW?**

These sentence transformers therefore depict a sentence in terms of one high-dimensional vector of semantic features. Comparisons between vectors of different sentences are therefore comparisons of the values for each encoded feature of a sentence. Predicated on the fact that these feature spaces are sufficiently representative of semantic meaning, this equates to comparing the semantic meaning of two sentences.

From linear algebra, a method of compounding the distance between two vectors is the cosine distance - functionally, this measures the direction between two vectors, and is derived from their dot-product (adjusting for magnitudes). This solution provides a meaningful comparison of the semantic similarity between two sentences as a number ranging from -1 to 1 .

2.7.2 Cross-Domain Applications

2.8 GitHub Popularity Metrics

2.8.1 Metrics as Proxies

2.8.2 Empirical Studies

2.9 Synthesis and Research Gap

2.9.1 Integrated Understanding

2.9.2 Remaining Challenges

2.9.3 Justification of Current Approach

2.10 Synthesis and Key Insights

- Labour demand forecasting methods exist, but few integrate innovation signals.
- OSS (especially GitHub) provides timely, innovation-related signals that may precede job market shifts.
- Job postings remain the most direct, scalable proxy for demand, but require careful NLP-based extraction.
- Embeddings enable cross-domain linkage between innovation (OSS) and demand (jobs), though validation is needed.
- Stars are an imperfect but widely used proxy for OSS adoption/popularity.

Chapter 3

Data Methods

3.1 Collection

In the case of this project, there are two facets of required data:

1. Open-Source Software Proliferation Data
2. Job Postings Data

3.1.1 Open-Source Software Data

Thankfully, the process of collecting data related to open-source software is fairly straightforward. As the name suggests, the OSS projects are intended to be *open* and accessible. This necessitates the usage of some online platform for hosting and maintaining the codebase, with strong version-control systems to safeguard a maintainable codebase. GitHub is functionally the default choice for hosting both open-source (and closed-source) codebases for a number of reasons, and is therefore the most relevant source for the required data of this project.

This project seeks data that more or less speaks to how *popular* a given GitHub repository is, and the prime candidate for this is the *GitHub Stars* mechanism. Users will "star" a repository in the same manner that internet users "bookmark" a website, behaving as a shortcut. This is a grossly simplified view of how GitHub repositories are interacted with, however - most GitHub repositories have no stars, and most users have "starred" very few repositories in return. The star is instead a useful proxy for measuring interest from a development point of view - if someone is contributing regularly to a repository, they will star it. In this sense, stars are a useful measure of how a repository is gaining interest from developers (and in a lesser sense, users), which in turn represents a growth in the utility the repository represents.

[11] is an open-source project that provides data on the historic star-counts of GitHub repositories, marketed as a tool to "provide insights into a repository's trendiness". Within the ML/AI framework that I delineated, the relevant repositories' star counts were found and downloaded thusly.

3.1.2 Job Posting Data

As is generally true of commercially valuable data, it can prove difficult to find open data that properly satisfies the desired criteria. Generally, it is a relatively simple task to build legally compliant scrapers that programmatically access and store the data from job boards online. However, finding stores of historical data presents a major challenge, because only *active* job postings are present on job boards. This means that the scraper must be active throughout and is not useful for collecting data in retrospect.

An alternative solution is to therefore find and access a pre-prepared dataset. This, unfortunately, proved challenging in the sense that:

- Large, high-quality datasets are presented as products, and are thus expensive
- Small, low-quality datasets often lack essential features, such as "job description" and even "date"

The solution to these problems was to utilize *several* datasets that occupy different time-regions, made available on Kaggle. An overview of the characteristics for each dataset is included in the Appendix, as per Table A.1.

At this rate,

Chapter 4

Representation

Chapter 5

Cross-Domain Similarity

Chapter 6

Time Series Methods

Chapter 7

Experiments

Chapter 8

Discussion of Results

Chapter 9

Critical Evaluation

In this chapter (it may not be Chapter 4 for you, but probably Chapter 6 or 7 once all the core chapters have been added.)

The critical evaluation consists of a discussion, leading to conclusion. It is an essential part of a master's degree.

It shows that you can not only carry out a substantial piece of work, but that you can reflect on it, and think critically about how you might have done it better.

Examiners view the critical evaluation as very important.

Critical evaluation should contain

- Strengths and weaknesses of your project
- If you were unable to attain any deliverables, then why
- What are the future plans for your project if you are to continue

You will be presenting this during demonstration but here you need to discuss them in details.

Chapter 10

Conclusion

A brief summary of all that has gone before.

May include some directions for future work.

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Appendix A

Generative AI

1. However I use Generative AI.

Data Sources

Dataset Name	Date Range	Details	Source
1000 ML Jobs US	12/2023 - 04/2025	Mostly within March & April of 2025, all jobs are ML related	[12]
Linkedin Data Engineer Job Postings	17/12/2023 17/12/2023	- Extra "skills" column - This is appended to "description"	[13]
Data Science Job Postings & Skills (2024)	19/01/2024 21/01/2024	- Extra "skills" column - This is appended to "description"	[14]
LinkedIn Job Postings (2023 - 2024)	23/03/2024 20/04/2024	- -	[15]
Job Postings from Ireland (October 2022)	01/10/2022 31/10/2022	- Ireland only; Sample dataset from Techmap.io's commercial product	[16]
US Job Postings from 2023-05-05	05/05/2023 05/05/2023	- USA Only; Sample dataset from Techmap.io's commercial product	[17]

Table A.1: Overview of datasets used for Job Postings

Third Party Code and Software Libraries

1. Whatever Software I end up using

A.0.1 ESCO Skill Classification

This publication uses the ESCO classification of the European Commission.

10.1 Word Counts

Category	Count
Words in text	3230
Words in headers	162
Words in captions	14
Tables/Figures	3
Inline math	9
Displayed math	0

Table 10.2: Document summary

File	Text	Headers	Captions	Floats	Inline Math	Displayed Math
report	0	16	0	0	0	0
./envs/argument-environment	121	0	0	0	0	0
abstract	39	0	0	0	0	0
preface	171	0	0	0	0	0
introduction	763	9	0	0	0	0
literature	1309	100	0	0	5	0
core/1-data	450	9	0	0	0	0
core/2-representation	0	1	0	0	0	0
core/3-similarity	0	2	0	0	0	0
core/4-timeseries	0	3	0	0	0	0
core/5-experimentation	0	1	0	0	0	0
core/6-discussion	0	3	0	0	0	0
critical-evaluation	134	2	0	0	0	0
conclusion	16	1	0	0	0	0
appendices	21	13	0	0	0	0
./figures/argument	327	0	0	0	4	0

Table 10.3: Word count by file (`texcount`)